PRIYANKA BOLEM

MACHINE LEARNING ENGINEER

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SUMMARY

ML Engineer with 4+ years building and shipping production models over high-volume, real-time data. Depth in predictive and probabilistic modeling, uncertainty estimation, and large-scale feature engineering, paired with direct ownership of low-latency inference on AWS SageMaker, distributed pipelines in Spark, Databricks, and Airflow, and production monitoring for drift and model decay. MS in Computer Science; strong applied grounding in Python, SQL, PyTorch, TensorFlow, and experimentation at scale.

TECHNICAL SKILLS

Languages: Python, SQL, R

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Random Forest, Hugging Face Transformers, OpenCV, NLP, Vector Databases

Modeling & Optimization: Regression, Classification, Clustering, Neural Networks, Bayesian Optimization, Monte Carlo Simulation, Genetic Algorithms, Reinforcement Learning, Hyperparameter Tuning, Grid Search

Big Data & Pipelines: Apache Spark, Kafka, Airflow, Azure Databricks, ETL design, feature engineering, data cleaning

MLOps & Deployment: AWS SageMaker (Pipelines, Endpoints), MLflow, Docker, Kubernetes, FastAPI, GitHub Actions, CI/CD, CloudWatch monitoring, drift detection

Cloud Platforms: AWS (SageMaker, Lambda, EC2), Azure (ML Studio, Databricks)

Evaluation & Experimentation: A/B testing, cross-validation, ROC-AUC, precision / recall / F1, model calibration, performance monitoring

Generative AI & NLP: GPT-4, BERT, LLaMA, LangChain, LangGraph, RAG, prompt engineering, tokenization

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Pandas, NumPy, Statsmodels

PROFESSIONAL EXPERIENCE

ML Engineer | Truveta
Seattle, WA | Jun 2025 - Present

- Deployed end-to-end ML pipelines on **AWS SageMaker**, using SageMaker Pipelines for training automation, SageMaker Endpoints for **real-time risk scoring**, and CloudWatch for production monitoring and alerting, ensuring reproducibility and compliance with healthcare data standards.
- Designed and implemented predictive models in Python and TensorFlow to forecast chronic disease progression from de-identified EHR and demographic data, enabling automated patient stratification for clinical research and population health studies.
- Applied **Bayesian Optimization and Monte Carlo simulation** across large patient cohorts to quantify uncertainty in outcome predictions, improving prediction precision and providing data-driven insight into intervention effectiveness.
- Built a hybrid optimization framework pairing Genetic Algorithms for cohort selection with Reinforcement Learning for adapting longitudinal strategies, integrating historical records with real-time signals for continuous learning; improved plan efficiency and outcome prediction accuracy by **22%**.
- Encoded high-cardinality unstructured text (lab reports, discharge summaries, medication history) as BERT embeddings, enriching model inputs with contextual signal from raw clinical language.
- Optimized SQL queries and data pipelines over large-scale datasets in a cloud-native data lake, reducing model training and data preparation time by **35%** and accelerating delivery for downstream data science and research teams.
- Partnered with research and data product teams to build Power BI dashboards tracking incidence trends, treatment efficacy across subpopulations, and regional risk distribution, supporting resource allocation across health systems.

ML Engineer | Capgemini
India | Apr 2019 - Nov 2022

- Built and deployed models with Scikit-learn, XGBoost, and LightGBM to predict credit default probability, loan delinquency, and fraudulent claims, then embedded them into origination, collections, and fraud triage systems with risk and compliance partners, reducing end-to-end loan decision cycle time by **25%**.
- Engineered features from large-scale financial, credit bureau, and behavioral data using Pandas and NumPy, capturing payment patterns, credit utilization, delinquency history, and spending trends.

- Built automated ETL pipelines in Python and SQL across core banking systems, credit bureaus, and CRM platforms, reducing data preparation time by **40%**; scaled preprocessing, feature engineering, and distributed training on Azure Databricks.
- Orchestrated **Apache Airflow** workflows for scheduled retraining, data refresh, and batch inference, keeping predictions accurate as customer behavior and regulatory requirements changed.
- Developed a **model performance monitoring system** in Python tracking accuracy, drift, and alert thresholds, reducing false positives by **20%** and improving early warning signals across credit and fraud models.
- Designed interactive Tableau dashboards for credit risk analysts, fraud teams, and loan officers to monitor approval rates, default probabilities, and fraud detection rates across customer segments, improving reporting efficiency by **18%**.

PROJECTS

- **Financial Sentiment MLOps Platform.** End-to-end NLP pipeline covering training, evaluation, deployment, and monitoring, with a live interactive demo hosted on Hugging Face Spaces.
- **Plant Disease Detection.** Deep learning image classification model built with TensorFlow and OpenCV to identify plant diseases from leaf imagery.
- **RAG Chatbot.** Retrieval augmented generation assistant combining vector search over a document corpus with an LLM response layer.
- **Lending Club Loan Approval.** Credit risk classification model on public lending data, covering feature engineering, class imbalance handling, and threshold tuning.
- Additional projects available at github.com/priyankabolem. Academic team projects (Attendance Buddy, Online Learning Platform) included ownership of system architecture and full stack development.

EDUCATION

Master of Science, Computer Science | Northwest Missouri State University, MO *May 2025*